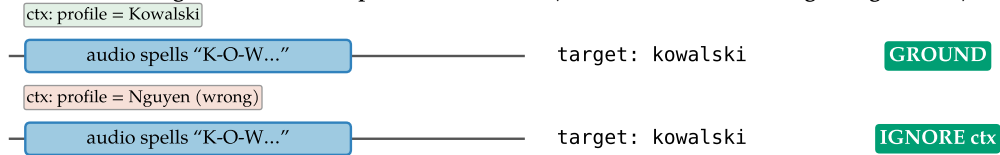


**Temporal axis: target must not depend on the future** (fix: same utterance  $\pm$  observable silence)



**Contextual axis: target must not be copied from context** (fix: same audio  $\pm$  a disagreeing context)



naive data (context always matches)  $\Rightarrow$  model copies context  $\Rightarrow$  40% wrong-profile intrusion; the counterfactual twin removes it